

6G Weekly Survey — Sep 14 - Sep 20, 2026

13 items across 5 topics, ranked by importance.

Physical AI

A Mathematical Theory of Pragmatic Information

Kai Niu, Ping Zhang · arXiv eess.SY · 2026-09-10

★★★★★ This is landmark theoretical work from Kai Niu (co-inventor of Polar codes) that provides the foundational mathematics for semantic/task-oriented communication, directly addressing 3GPP's emerging 6G vision of goal-oriented networks and AI-native air interfaces.

This paper proposes a pragmatic information theory that extends Shannon's framework by formalizing how information drives goal-directed actions rather than just symbol fidelity. It introduces isoteleia mappings (semantic paths yielding equivalent actions) and defines pragmatic entropy, mutual information, and channel capacity, proving three coding theorems that generalize classical results. The framework includes pragmatic value/cost of information metrics and establishes fundamental capacity bounds for resource-constrained intelligent systems performing decision-making tasks.

Contribution: The isoteleia mapping formalism and three-tier syntactic-semantic-pragmatic hierarchy rigorously unify communication and control theory, yielding pragmatic coding theorems and a behavioral capacity limit that generalizes Shannon capacity from symbol transmission to goal-directed action.

Relevance to your work: This framework is central to your semantic and task-oriented communication research, providing rigorous mathematical foundations for designing AI-native PHY layers where transmission is optimized for downstream decision-making rather than waveform reconstruction.

ISAC (sensing + communication)

CORDIS: A Scalable Coordinated Resource Allocation Framework for Distributed Cell-Free ISAC

Mehdi Zafari, Björn Ottersten, A. Lee Swindlehurst · arXiv eess.SY · 2026-09-10

★★★★■ Strong contribution addressing a critical scalability gap in multi-static ISAC deployments with practical constraints (imperfect CSI, fronthaul limits), from a credible research group with rigorous optimization methodology. This paper addresses coordinated resource allocation in distributed cell-free ISAC networks, where centralized solutions with full CSI are impractical. It proposes CORDIS, a framework with two algorithms: CORDIS-Split (fixed local beamformers plus centralized power allocation) and CORDIS-ADMM (consensus ADMM for joint beamforming and power optimization). Both ensure fronthaul overhead and per-AP computation scale independently of antenna/AP counts, with CORDIS-ADMM approaching centralized performance while degrading gracefully under imperfect CSI.

Contribution: First scalable distributed optimization framework for cell-free ISAC that decouples computational complexity from network scale and handles imperfect CSI robustly through localized matrix operations and consensus ADMM.

Relevance to your work: Directly relevant to your ISAC interests, particularly for distributed architectures where coordination overhead and robustness to channel estimation errors are fundamental challenges in realizing practical joint sensing-communication systems.

Semi-Cooperative Passive Integrated Sensing and Communication by Utilizing Physical Layer Information of 5G Signals

Bo Wei, Ryusei Ogane, Hang Song · arXiv eess.SY · 2026-09-09

★★★ ■ Solid practical contribution addressing real deployment constraints for passive ISAC, though the sensing algorithms themselves appear standard and the experimental validation is preliminary (multipath detection is basic).

This paper implements a semi-cooperative passive ISAC system using open-source 5G framework (likely srsRAN or OpenAirInterface) and software-defined radios to extract physical layer information from 5G signals for sensing. The authors experimentally evaluate sensing performance across multiple communication scenarios (synchronization, pinging, data transmission) with variable bandwidths and successfully detect multipath propagation using two different approaches. The work addresses the practical challenge that commercial cellular systems do not expose protocol stack details needed for passive sensing research.

Contribution: Demonstrates a practical experimental platform bridging the gap between commercial 5G deployments and ISAC research by using open-source stacks to extract PHY-layer information for passive sensing without requiring full base station cooperation.

Relevance to your work: Directly relevant to your ISAC work as it provides a reproducible methodology for passive sensing experiments with actual 5G waveforms, which could inform your own testbed development or semi-cooperative ISAC algorithm design where full network access is unavailable.

NTN / satellite

Adaptive Distributed Physical-Layer Authentication and Attack Detection in 6G Non-Terrestrial Networks via Causal Meta-Learning

Parsa Rajabi, Mohammad Reza Abedi, Nader Mokari, Paeiz Azmi, et al. · arXiv eess.SY · 2026-09-08

★★★★■ Addresses a critical 6G security challenge with a principled causal ML approach and demonstrates strong empirical results on NTN-specific distribution shifts, though lacks real-world validation or standardization pathway. This paper addresses physical-layer authentication (PLA) in non-terrestrial networks where Doppler shifts, delays, and rapid channel variations cause distribution shifts that degrade conventional learning. The authors propose SAFA-MZ, a causal meta-learning framework using MAML with invariant risk minimization that fuses multi-feature fingerprints (spatial, angular, Doppler-delay) from distributed aerial nodes and employs a two-stage authentication with selective TDOA-based graph attention network localization. Simulations demonstrate 92% accuracy and 96% AUC, outperforming centralized deep learning baselines across diverse NTN environments with fast adaptation using few labeled samples.

Contribution: Novel combination of structural causal modeling with meta-learning (MAML+IRM) for distributed physical-layer authentication that achieves fast adaptation to unseen NTN environments through multi-feature fusion and causal consistency regularization.

Relevance to your work: Directly relevant to your NTN research interests; the meta-learning framework for handling NTN channel dynamics and the multi-feature fusion approach could inform your work on AI-native air interfaces for non-terrestrial systems.

Distributed Physical Layer Authentication and Collaborative RSMA in Non-Terrestrial Networks via Graph Reinforcement Learning

Parsa Rajabi, Mohammad Mirzaee, Mohammad Reza Abedi, Nader Mokari, et al. · arXiv eess.SY · 2026-09-08

★★★★■ Solid incremental work combining known techniques (PLA, RSMA, graph RL) for NTN security, but lacks experimental validation or channel modeling insights relevant to 6G standardization.

This paper proposes a distributed physical-layer authentication scheme for non-terrestrial networks that embeds group-level authentication tags into rate-splitting multiple access (RSMA) transmissions while protecting against eavesdropping. The method jointly optimizes HAPS placement, user association, and RSMA power allocation using graph-aware actor-critic reinforcement learning (GA2C) and a repair-based cross-entropy method. Simulations show 135% secrecy spectral efficiency gain over single-connect baselines and 21% over schemes without artificial noise.

Contribution: First joint design of distributed physical-layer authentication with collaborative RSMA in NTNs, incorporating group differential privacy for tag protection and graph RL for scalable optimization.

Relevance to your work: Relevant as an NTN security study using RL for resource allocation, though the focus on authentication and differential privacy differs from your core interests in AI-native PHY, channel

modeling, and ISAC.

Tethered UAVs for Dense Urban Connectivity

German Svistunov, Azim Akhtarshenas · arXiv eess.SY · 2026-09-09

★☆☆☆☆ Competent system-level study of tethered UAVs but offers narrow practical scope—tethering limits mobility and scalability, and analysis remains conventional SINR/throughput metrics without novel physical-layer or AI techniques. This paper compares tethered UAV base stations (TUAVs) against terrestrial 5G in dense urban scenarios using 3GPP-compliant system-level simulations with 19 sites and 57 sectors. TUAVs avoid battery constraints and enable stable positioning at altitudes from 100–1000 m, achieving near-LOS links. Results show TUAVs significantly improve cell-edge throughput when altitude is optimized to balance LOS probability against propagation loss and interference.

Contribution: First 3GPP-compliant multi-cell performance evaluation of tethered (not battery-limited) UAV base stations in realistic dense urban topology, demonstrating altitude-dependent tradeoffs between LOS gains and interference.

Relevance to your work: tangential

6G Demos & Trials

Mitigating Degradation Attacks in Cooperative Autonomous Driving via Intention Sharing: A Vehicle-in-the-Loop Study

Prakhar Gupta, Tyler Ard, Rongyao Wang, Jagruti Sahoo, et al. · arXiv eess.SY · 2026-09-09

★☆☆☆☆ Competent experimental work on vehicular resilience, but focuses on control-layer mitigation rather than addressing fundamental wireless/PHY vulnerabilities relevant to 6G air interface design.

This paper investigates cooperative autonomous driving under Denial-of-Service attacks causing up to 2-second communication delays, using a vehicle-in-the-loop testbed with real drive-by-wire hardware. They compare baseline status-sharing controllers against intention-sharing and delay-aware intention-sharing variants under model predictive control. Results show intention-sharing eliminates collisions and maintains near-nominal performance despite adversarial delays, while baseline control suffers degradation and frequent collisions.

Contribution: Vehicle-in-the-loop experimental validation that intention-sharing communication architectures can maintain safe cooperative driving performance under realistic DoS-induced V2X delays up to 2 seconds.

Relevance to your work: tangential

LettuceVisSim: A Simulator That Generates Lettuce Image Time-series for Vision-Based Reinforcement Learning

Ziye Zhu, Bert van 't Ooster, Congcong Sun, Eldert van Henten, et al. · arXiv eess.SY · 2026-09-11

★☆☆☆☆ While technically competent for agricultural simulation, this work is entirely orthogonal to wireless communications, physical layer design, or any aspect of 6G systems development.

The paper presents LettuceVisSim, a simulator for generating labeled time-series images of lettuce growth to enable vision-based reinforcement learning in controlled environment agriculture. The system combines a process-based biomass model, canopy layout algorithm, and Unity rendering engine to produce RGB and segmentation images in <10ms. Validation shows $R^2=0.84$ for shoot dry weight prediction and $R^2=0.84$ for ground coverage ratio, with a proof-of-concept demonstrating learned lighting-control policies from simulated images.

Contribution: First validated simulator capable of generating labeled crop image time-series at high speed for training vision-based RL agents in agricultural decision-making, addressing the scarcity of labeled agricultural training data.

Relevance to your work: tangential

Transducer Placement and the Limits of a Four-State Reduced Model in Post-Flutter Piezoelectric Energy Harvesting from a Pitch-Plunge-Flap Aerofoil

Nikolaos D. Tantaroudas, Ilias Karachalios, Andrew J. McCracken · arXiv eess.SY · 2026-09-11

★☆☆☆☆ This is a specialized aeroelastic energy harvesting study with no connection to wireless communications, physical layer design, or any aspect of 6G systems.

This paper studies piezoelectric energy harvesting from aeroelastic flutter in a pitch-plunge-flap airfoil system, analyzing how transducer placement affects both the flutter boundary and harvested power. A fifteen-state electro-aeroelastic model is reduced via biorthonormal projection, revealing that reduced-order model accuracy depends critically on capturing cross-modal coupling (via reprojecting the exact Jacobian) rather than simply expanding the retained basis. The authors find that harvester voltage prediction errors persist even when electrical modes are included, demonstrating that reduced models must be validated on the harvested quantity itself, not just structural dynamics.

Contribution: The work identifies that frozen-basis reduced-order models fail to predict harvested voltage accurately because equilibrium eigenvectors lock voltage in fixed proportion to motion, and shows that reprojecting the exact Jacobian (rather than expanding eigenvalues) is necessary to recover structural response without basis enlargement.

Relevance to your work: tangential

Distributed Optimization of Modular Production Systems using Model-based Reinforcement Learning with Inverse Models

Andreas Schwung, Steve Yuwono, Sofiene Lassoued, Dorothea Schwung · arXiv eess.SY · 2026-09-10

★☆☆☆☆ This work addresses industrial automation and manufacturing optimization, which is orthogonal to wireless physical layer and 6G air interface research.

The paper proposes a model-based reinforcement learning framework for controlling modular manufacturing systems by integrating approximate inverse process models into the policy network. This approach separates actuation dynamics from state-space dynamics, enabling RL training solely in task space. Experiments on a laboratory testbed demonstrate improved performance and faster training, especially for off-policy algorithms.

Contribution: The novel integration of lightweight feedforward inverse models within RL policy networks to decouple actuation and state-space learning for modular manufacturing control.

Relevance to your work: tangential

GeoTrussRover: Morphological Computation with Contact-Semantic Control Primitives

Muyuan Ma, Yi Zhang, Yang Yang, Xuanyan Zheng, et al. · arXiv eess.SY · 2026-09-10

★☆☆☆☆ This work addresses terrestrial mobile robotics and mechanical reconfiguration control, which has no direct relevance to wireless communication systems, physical-layer design, or 6G network architectures.

This paper presents GeoTrussRover, a reconfigurable robot with a variable-geometry truss structure that adapts its morphology for obstacle traversal. The authors develop contact-semantic control primitives that coordinate 21 truss members and wheels, enabling efficient motion adaptation across step heights (1.08–4.97 wheel radii) by selectively recomputing only infeasible phases rather than full trajectories. An electric prototype validates traversal up to 2.11 wheel radii with 63.7% reduction in optimization evaluations compared to full recomputation.

Contribution: The introduction of contact-semantic primitives that enable efficient morphology adaptation through physics-constrained projection and selective phase recomputation, reducing computational cost while maintaining feasibility across a wide range of obstacle geometries.

Relevance to your work: tangential

Economic Evaluation of V2G-Enabled Fast Charging Stations Under Endogenous EV Adoption Dynamics

Mingjian Tuo, Jie Zhou, Yao Yan, Cunzhi Zhao, et al. · arXiv eess.SY · 2026-09-09

★☆☆☆☆ This is competent power systems optimization work but completely outside wireless communications research—it addresses electric vehicle charging economics and power grid planning with no connection to physical layer, channel modeling, or radio access technologies.

This paper develops a multi-period planning framework for vehicle-to-grid (V2G)-enabled fast charging stations that explicitly models the feedback loop between charging infrastructure deployment and electric vehicle adoption rates. The authors introduce an aggregated fleet virtual battery model to represent V2G flexibility at scale and use SOS2 variables with SOCP relaxations to make the mixed-integer optimization tractable. Numerical experiments on coupled transportation and power distribution networks demonstrate improved social welfare and reduced grid congestion through macroscopic V2G aggregation.

Contribution: Treats EV adoption as decision-dependent uncertainty in infrastructure planning and introduces a computationally efficient macroscopic V2G fleet model that avoids tracking individual vehicle state-of-charge.

Relevance to your work: tangential

Verticals & use cases (XR, digital twins, IIoT)

High-Fidelity Multi-Body Simulator for Autonomous Racing

Nicola Musiu, Francesco Iacovacci, Fausto Lupo, Matteo Pini, et al. · arXiv eess.SY · 2026-09-11

★★☆☆☆ While technically competent, this is a vertical-specific simulation tool for autonomous racing with limited direct impact on wireless PHY research or 6G system design.

The authors develop a high-fidelity multi-body vehicle dynamics simulator for autonomous racing by creating a digital twin in Dymola, exporting it as a Functional Mock-up Unit (FMU), and integrating it into a software-in-the-loop testing framework with real-time performance on portable hardware. The simulator incorporates detailed 3D road surfaces using the Curved Regular Grid standard and is calibrated/validated against experimental racing data. It enables testing of autonomous racing algorithms before real-world deployment.

Contribution: A validated real-time digital twin framework for autonomous racing that bridges high-fidelity multi-body dynamics modeling with practical software-in-the-loop testing on commodity hardware.

Relevance to your work: Tangential—the digital twin concept relates to XR/industrial IoT use cases requiring ultra-low latency wireless, but this work focuses entirely on vehicle dynamics simulation without addressing wireless communication aspects.